



Materials removal and adhesive bonding joint strengthening mechanisms of aluminum-based silicon carbide metal matrix composites by femtosecond laser processing



Jiecai Feng ^a  , Hengyi Song ^a, Xing Wang ^b, Hongfei Liu ^c, Yanning Sun ^a, Xiaohai Peng ^d, Yilian Zhang ^a, Yingzhong Tian ^a

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<https://doi.org/10.1016/j.ceramint.2025.12.509> 

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Abstract

Aluminum-based silicon carbide particles metal matrix composites (SiCp/Al composites) is widely applied in aerospace due to its excellent properties. Adhesive bonding is the common connection method for SiCp/Al composites. How to carry out efficient and high-quality surface treatment of the SiCp/Al composites before the adhesive bonding is a key factor to improving the mechanical properties. Recently, with ultra-high instantaneous energy, the femtosecond laser is a feasible surface treatment method for the SiCp/Al composites. However, the materials removal and adhesive bonding joint strengthening mechanisms of the SiCp/Al composites by femtosecond laser processing are still unclear. Thus, in this paper we calculated the ablation threshold of SiCp/Al

composites by femtosecond laser processing based on the diameter regression method and analyzed the morphology and composition of the micropores ablated by the femtosecond laser processing. The tensile strength, elongation and surface morphology of fracture section of the adhesive bonding joints with micropores array were also investigated in details. The results showed that when the laser pulse energy was equal or less than $10\mu\text{J}$, periodic microstructure of multi-layer fish scales were obtained in the inner wall of the micropore due to the non-thermal coupling effect of coulomb explosion, vaporization and plasma shock. When the laser pulse energy exceeded $10\mu\text{J}$, brittle fracture steps were obtained at the edge of the volcanic crater-shaped micropore mainly due to the recoil pressure from plasma expansion. The results also indicated that the mechanical properties of the adhesive bonding joints with micropores array had been significantly improved due to the pinning effect with mechanical interlocking mechanism.



Keywords

Aluminum-based silicon carbide metal matrix composites; Femtosecond laser; Laser processing; Materials removal; Adhesive bonding joint; Microstructure

1. Introduction

Aluminum-based silicon carbide metal matrix composites (SiCp/Al composites) are one of the important metal matrix composites, found many applications like of aerospace [1] and automobile [2] due to its lightweight, excellent thermal stability and high specific strength [3]. Adhesive bonding is increasingly popular method for joining SiCp/Al composites due to its effectiveness and economy over mechanical fasteners method [4].

Surface morphology of the adhesive bonding joint is a key factor for controlling the mechanical properties of the joint. [[5], [6], [7]]. The surface morphology can be optimized by various surface treatment method such as micro-milling [8], sandblasting [9], etching [10] and laser ablation [[11], [12], [13]]. Among the those surface treatment methods, laser ablation is more suitable for

the surface treatment of SiCp/Al composites, due to its advantages such as low deformation, high efficiency and no waste liquid pollution [14,15].

However, the materials removal mechanism of the SiCp/Al composites by femtosecond laser processing is still unclear [[16], [17], [18]]. Zhang et al. [19] confirmed that the existence of laser-induced periodic surface structures LIPSS enabled the removal of materials by redistributing the surface light intensity when the laser pulse energy was below the gasification threshold. Once the laser pulse energy reached the gasification threshold, the removal of materials will occur more rapidly in the form of vaporization. Zhou et al. [20] presented that in the investigation of removal mechanism of SiC composites by femtosecond laser processing, the heat accumulation effect and oxidation degree were increased accordingly as the laser pulse energy increased and that the materials removal mechanism by the femtosecond laser transferred from non-thermal removal mechanism to the equilibrium stage mechanism and then to thermal removal mechanism. They also reported that the oxidation behavior of SiC was complicated. When the gradual enhance of the laser pulse energy, the dominant oxidation mechanism gradually transferred from passive oxidation of SiC to active oxidation of SiC.

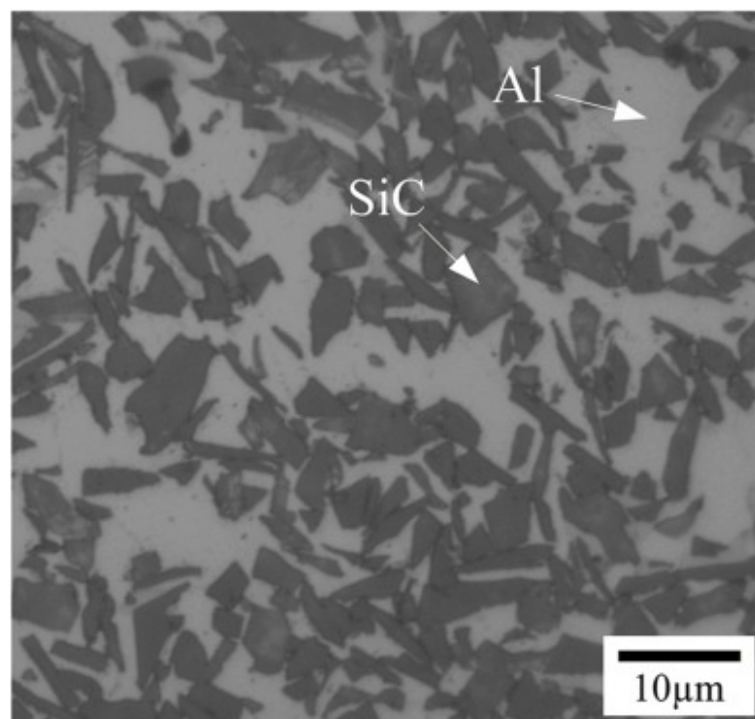
Thus, this paper focus on revealing the materials removal SiCp/Al composites by the femtosecond laser ablation and analyzing the adhesive bonding joints strengthening mechanisms of SiCp/Al composites by using a three dimensions laser scanning confocal microscope (LSCM), a scanning electron microscopy (SEM), and an energy dispersive spectrometer (EDS). The ablation threshold of SiCp/Al composites by femtosecond laser processing based on the diameter regression method was calculated, and the morphology and composition of the micropores ablated by the femtosecond laser processing were analyzed. Additionally, the tensile strength, elongation and surface morphology of fracture section of the adhesive bonding joints with micropores array were also investigated in details.

2. Methods

2.1. Materials

The SiC_p/Al composites used in this study was 2009 Al metal matrix and 45% of SiC particles by volume (Fig. 1). The diameter of the SiC particles was about 10μm. The chemical composition of the composites is shown in Table 1. The specimens with dimensions of 100mm×25mm×3mm were ultrasonically cleaned in anhydrous ethanol for about 5min before the femtosecond

laser processing. The adhesive used in this study was A420 mixed adhesive. The parameters of the A420 mixed adhesive are shown in Table 2.



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Fig. 1. Morphology of SiC_p/Al composites.

Table 1. Chemical composition of SiC_p/Al composites (wt.%).

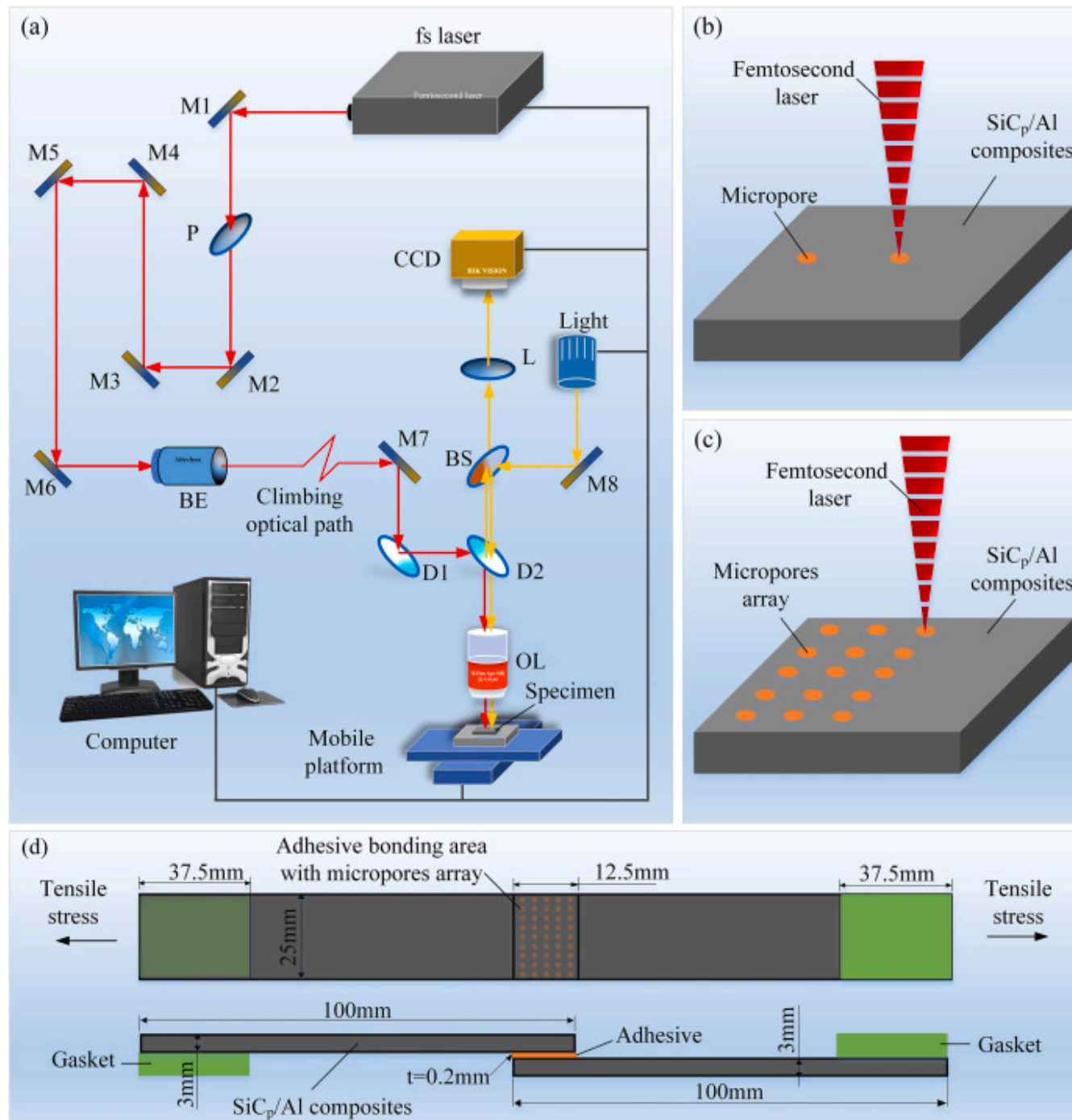
Al	Si	C	Cu	Mg	Mn	Zr	Zn	Fe
46	35.9	15.3	2	0.25	0.2	0.1	≤0.15	≤0.1

Table 2. Parameters of the A420 mixed adhesive (25°C).

Density (g/cm ³)	Viscosity (Pa·s)	Initial solidification time (h)	Complete solidification time (h)	Glass transition temperature (°C)	Temperature tolerance range (°C)
1.1	35–45	6	48	75–80	–55~120

2.2. Femtosecond laser processing system and adhesive bonding

[Fig. 2 \(a\)](#) presents the schematic diagram of the femtosecond laser processing system for the SiCp/Al composites. [Fig. 2 \(b\)](#) presents the schematic diagram of the micropore by femtosecond laser processing of the SiCp/Al composites. [Fig. 2 \(c\)](#) displays the schematic diagram of laser ablation of the micropores array.



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Fig. 2. Schematic diagram of: **(a)** femtosecond laser processing system, **(b)** micropore ablated by femtosecond laser processing, **(c)** micropores array fabrication and **(d)** adhesive bonding joints tensile test.

The system mainly constituted of a femtosecond laser (Femoto YL-40), some optical components, a light, a charge coupled device camera (CCD) and a computer. The pulse width of the femtosecond laser was 220 fs and the wavelength was 1030nm. The maximum average power, M^2 and maximum repetition frequency of the femtosecond laser were 50W, 1.16 and 5MHz, respectively. With burst mode, a single shot could be chosen in the femtosecond laser processing. A single-pulse mode was used in the micro-hole ablation experiments.

The path of the femtosecond laser was as follows: firstly, the femtosecond laser went through mirror 1 (M1), polarizer (P). Secondly, the laser went through from mirror 2 (M2) to mirror 6 (M6). Thirdly, the laser went through beam expander (BE), and climbed up with the climbing optical path. Finally, the laser went through mirror 7 (M7), dichroic mirror 1 (D1) and dichroic mirror (D2), focused objective lens (OL) and the specimen in sequence. A light with wavelength of 600nm was adopted for illumination. The path of the illumination light went through mirror 8 (M8), BS, D2, OL and the specimen. The path of the illumination light could reversely pass through the CCD through the OL, D2, BS and focus lens (L). A 10 times objective lens (Mitutoyo) was used. The radius of the focused laser spot was 6 μ m.

During the femtosecond laser processing for micropores experiment, in order to enhance the reliability of the test and reduce the deviation caused by accidental results, three repeated micropores ablation experiments were conducted with the same parameters. For the micropores ablation, the femtosecond laser focused on the surface of the specimens. The repetition frequency of the femtosecond laser was 100kHz. The laser pulse energies were from 1 μ m to 73.6 μ J. For the influence of repetition frequency, the average laser power was 3W and the repetition frequency were from 50kHz to 2500kHz. For the micropores array ablation, the laser pulse energies were 10 μ J, 30 μ J and 40 μ J. The repetition frequency was also 100kHz. The horizontal and vertical spacing of the array were both 0.35mm. The scanning velocity was 12.5mm/s.

Fig. 2 (d) exhibits the schematic diagram of adhesive bonding test process according to the standard of GB/T 7124-2008. The adhesive bonding area was 25mm $\times$ 12.5mm with micropores array. The thickness of the adhesive layer was 0.2mm. During the tensile test, the clamping length of the specimen was 37.5mm at both ends of the specimens. Firstly, the A420 adhesive was

applied on the surface of the corresponding adhesive area. Secondly, the actual adhesive bonding area needed to be measured to ensure the accuracy of the subsequent tensile strength calculation. Thirdly, the test samples were pressed and fixed for compression bonding in order to enhance the adhesion between the materials and the adhesive. Finally, with the gaskets on both sides of the specimens, the adhesive bonding joints were tested on a Meitesi tensile testing machine. The tensile test velocity was 0.5kN/s.

2.3. LSCM and SEM

The morphology of the ablation specimens and the failure surface morphology of the adhesive bonding joint were analyzed by using a Keyence VK-X3100 3D laser scanning confocal microscope (LSCM). The longitudinal and transverse resolution of the LSCM were 0.2μm and 0.1μm, respectively.

The microstructure and component of the specimens were analyzed by using a scanning electron microscopy (GeminiSEM) 300with EDS from ZEISS. Before SEM analyze, the specimens were plated treatment with thickness of about 5nmPt.

2.4. Simulation

The femtosecond laser processing of SiC_p/Al composites was simulated using the COMSOL software. The energy distribution in femtosecond pulse laser processing is uneven, generally showing the characteristic of concentration at the center and attenuation at the edges. This energy distribution is consistent with that of a Gaussian laser source. The distribution form of a Gaussian laser source conforms to the characteristics of a normal distribution, with the temperature at the center of the laser spot being the highest, and the temperature gradually decreasing as the distance from the center of the laser spot increases. Thus, it is reasonable to choose a Gaussian laser source in the simulation process. Equation (1) represents the energy density of a laser pulse that is Gaussian-distributed in space and time:

$$E(r) = \frac{P}{\pi f \omega^2} \cdot e^{-\frac{2r^2}{\omega^2}} \quad (1)$$

where, E is the laser pulse energy (J), P is the average laser power (W), f is the laser repetition frequency (Hz), r is the distance from any point on the laser processing plane to the center of the laser spot and ω is the radius of the laser spot.

Heat conduction equation in the Cartesian coordinate system:

$$\rho C_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + Q(x, y, z, t) \quad (2)$$

Where, ρ represents the density of SiC_p/Al composites, C_p is the specific heat capacity of SiC_p/Al composites, k is the thermal conductivity of SiC_p/Al composites, T is the temperature of SiC_p/Al composites, and Q is the heat absorbed during the processing.

The mathematical description of thermal convection usually employs the convection heat transfer equation:

$$Q_s = h(T_2 - T_1) \cdot A \quad (3)$$

where, h represents the convective heat transfer coefficient, T_2 is the temperature of the heat-releasing fluid (or solid), T_1 is the temperature of the heated fluid (or solid) and A is the heat exchange area.

The calculation of radiant heat can be expressed by formula (4):

$$Q_f = -\varepsilon \sigma (T^4 - T_0^4) \cdot A \quad (4)$$

where, ε represents the emissivity of the material surface, σ is the Stefan-Boltzmann constant with, $\sigma = 5.67 \times 10^{-8} \text{ W} / (\text{cm}^2 \cdot \text{K}^{-4})$.

To simulate the material removal process of vaporization caused by absorbing heat on the material surface, the "moving mesh" setting was used in COMSOL to model this process. The amount and speed of material removal on the surface are closely related to the absorbed heat. The materials ablation speed defined in COMSOL is as shown in formula (5):

$$v = \frac{q_a}{\rho H_s} \quad (5)$$

where, v represents the ablation rate of the materials, q_a is the heat carried away by materials ablation, ρ is the density of the materials and H_s is the sublimation heat of the materials.

3. Results and discussion

3.1. Ablation threshold

The ablation threshold (F_{th}) is one of the key parameters in the interaction process between the femtosecond laser and the materials [21]. The F_{th} presents to the minimum laser energy density at which the material begins to be ablated with specific laser energy density [22]. The calculation of F_{th} is usually based on experimental data and theoretical models [23].

The calculation of F_{th} requires first determining the theoretical spot beam waist radius. In this study, the objective lens used was LMH-10X-1064, with a numerical aperture (NA) of 0.25, and the aperture of the objective lens matched the incident laser beam of the laser. The theoretical spot beam waist radius was 6 μm . A Gaussian laser beam was used in this study. The energy density distribution of the Gaussian laser beam satisfy formula (6):

$$F(r) = F_0 \exp\left(\frac{-2r^2}{\omega_0^2}\right) \quad (6)$$

$$F_0 = \frac{2E}{\pi\omega_0^2} \quad (7)$$

$$E = \frac{P}{f} \quad (8)$$

where, F_0 represents the peak energy density at the center of the beam (J/cm^2), ω_0 is the beam waist radius (μm), and r is the distance from a point in the beam to the center of the beam (μm), E is the laser pulse energy (J), P is the average laser power (W), and f is the laser repetition frequency (Hz). Substituting Equation (8) into Equation (7) gives:

$$F_0 = \frac{2P}{f\pi\omega_0^2} \quad (9)$$

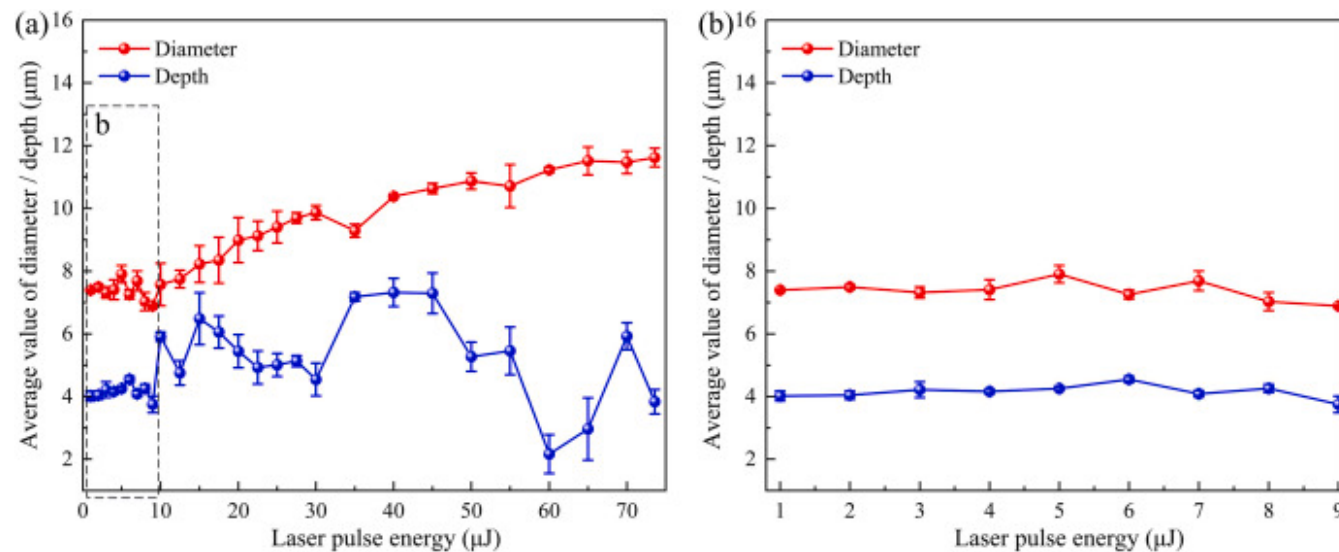
when the equivalent ablation diameter under the laser F_{th} condition reaches 0 μm , let D represents the diameter of the micropore after laser ablation. Combining Equations (6), (9), the following formulas were obtained:

$$F_{th} = F_0 \exp\left(\frac{-D^2}{2\omega_0^2}\right) \quad (10)$$

$$D^2 = 2\omega_0^2 \ln\left(\frac{F_0}{F_{th}}\right) \quad (11)$$

From Equation (11), by measuring the ablation micropore diameter D with different energy densities F_0 by experiments, a curve of D^2 versus $\ln(F_0)$ can be plotted. Thus, the ablation threshold F_{th} can be calculated by fitting a straight line.

To calculate the ablation threshold of SiCp/Al composite materials by the femtosecond laser processing, the diameters of ablation micropores were obtained, which were ablated by femtosecond laser processing with constant repetition frequency of 100kHz and different laser energy from 1 μ J to 73.6 μ J (Fig. 3). Fig. 3 presented that the average diameter of the micropores were ranged from 7 μ m to 12 μ m and the average depth of the micropores were between 2 μ m and 7 μ m. The results also demonstrated that when the laser pulse energy were 1 μ J–9 μ J, the average diameter of the micropores were no significantly changed (Fig. 3(b)), while the average diameter of the micropore enhanced with increasing of the laser pulse energy from 10 μ J to 73.6 μ J (Fig. 3(a)). However, the maximum average depth of the micropores was about 7 μ m, when the laser pulse energy was about 40 μ J. The minimum average depth of the micropores was about 2 μ m, when the laser pulse energy was about 60 μ J. When the laser pulse energy were 1 μ J–9 μ J, the average depth of the micropores were no significantly changed with value of about 4 μ m.

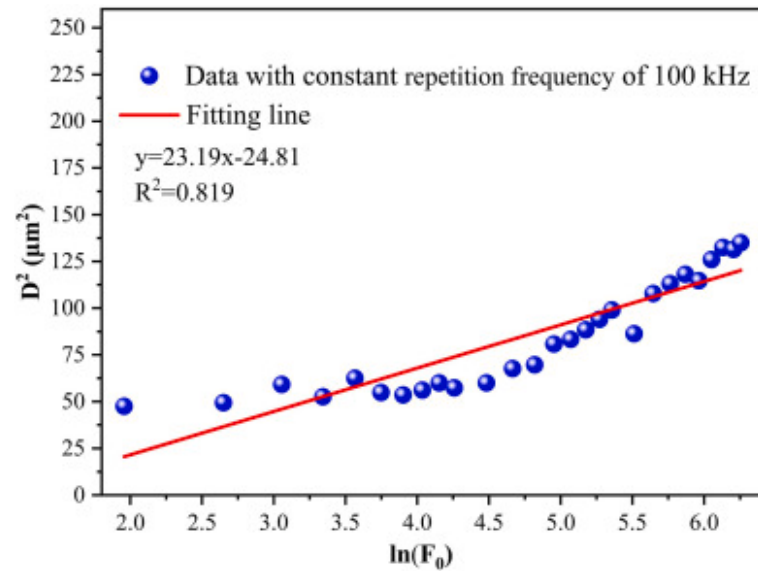


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Fig. 3. Average diameter and depth of the ablation micropores by femtosecond laser processing with constant repetition frequency of 100kHz: **(a)** value distribution with laser energy from 1 μ J to 73.6 μ J and **(b)** value distribution with laser energy from 1 μ J to 9 μ J.

A straight line obtained through data fitting is shown in Fig. 4. The results indicated that the single-pulse ablation threshold of SiCp/Al composite materials by the femtosecond laser processing is 2.91 J/cm² with a repetition frequency of 100kHz.



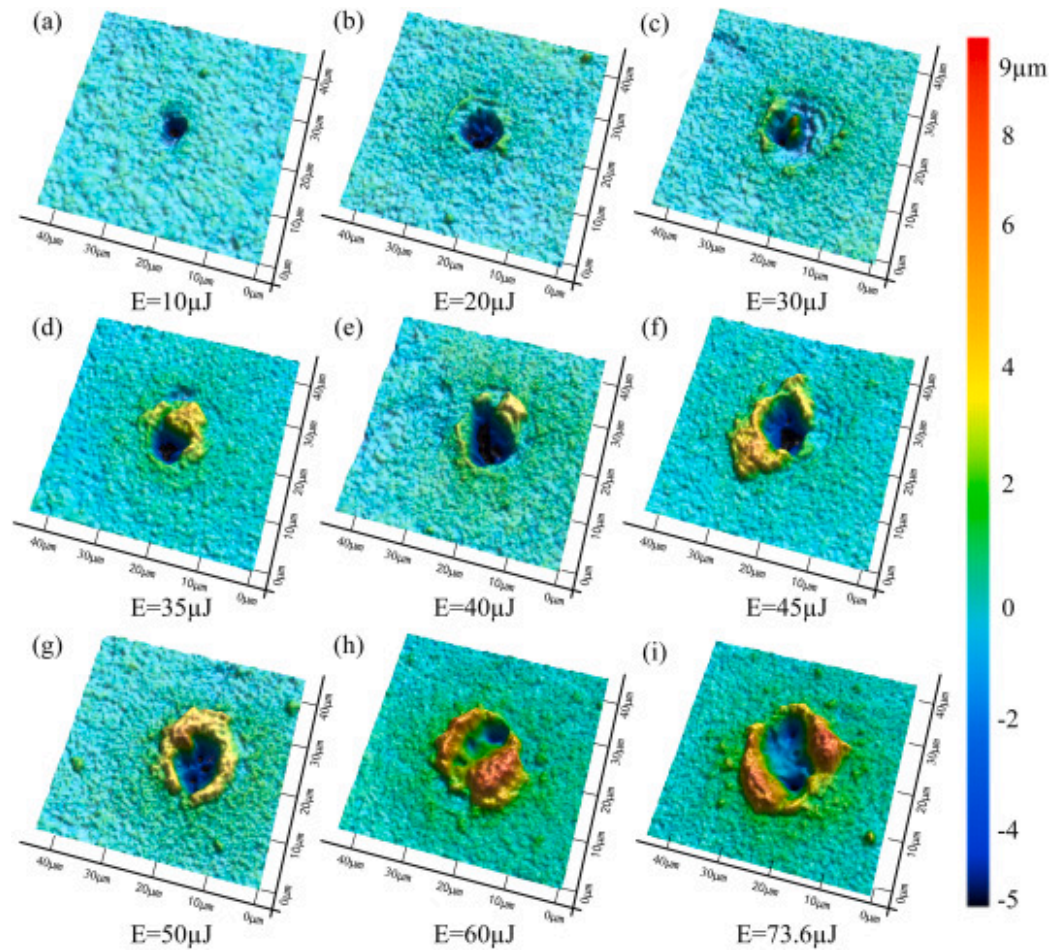
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Fig. 4. Fitting line between the square of diameters of the micropore and the logarithm of laser pulse energy with constant repetition frequency of 100kHz.

3.2. Materials removal mechanism

The three-dimensional morphology of the ablation micropores by femtosecond laser processing with different laser pulse energy (E) were presented in Fig. 5. On the right side of Fig. 5, the color bar is the dimension variation with value from $-5 \mu\text{m}$ to $9 \mu\text{m}$.



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Fig. 5. Three dimensions morphology of ablation micropores by femtosecond laser processing with different laser pulse energy (E): (a) 10 μ J, (b) 20 μ J, (c) 30 μ J, (d) 35 μ J, (e) 40 μ J, (f) 45 μ J, (g) 50 μ J, (h) 60 μ J and (i) 73.6 μ J.

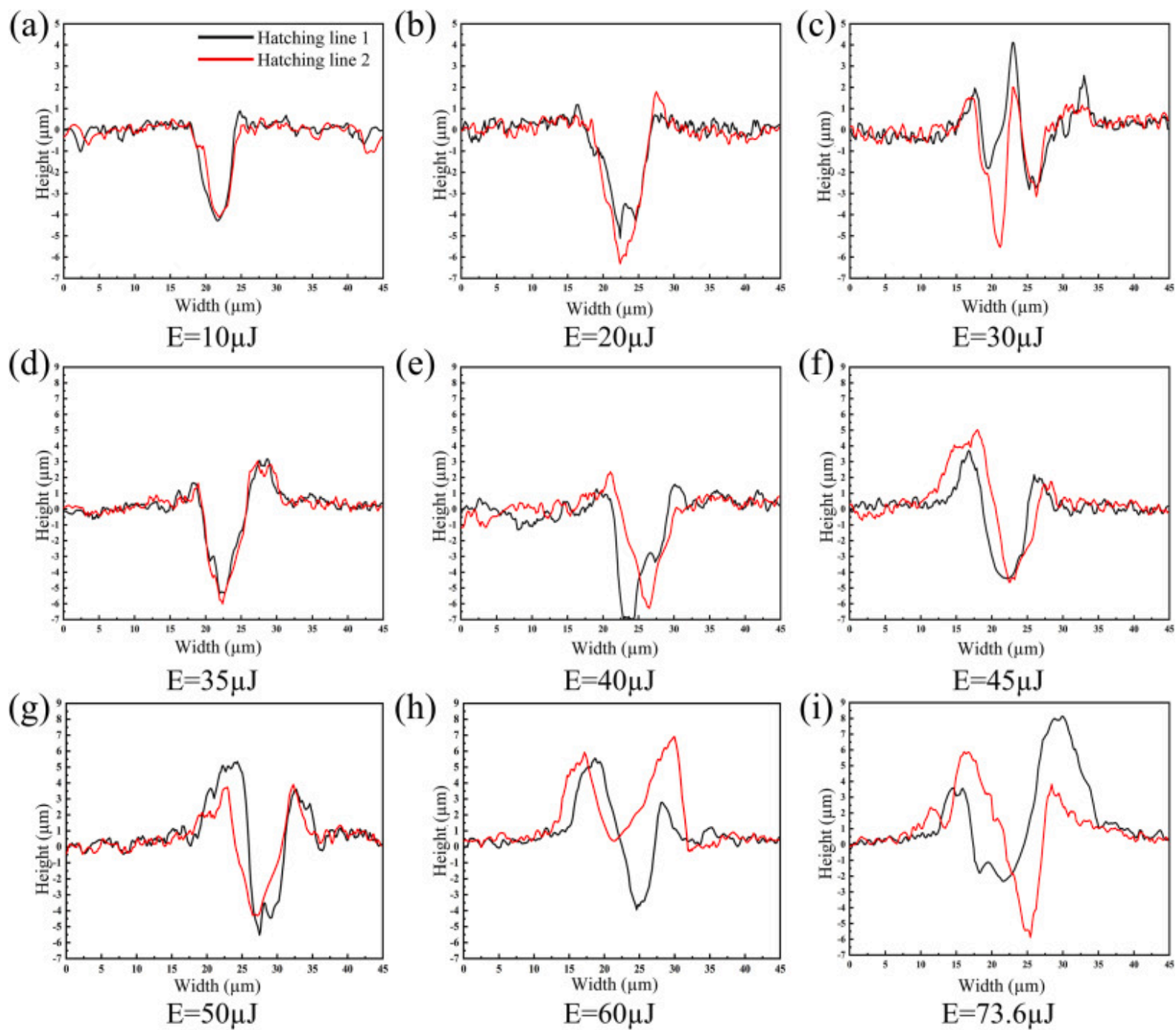
Fig. 5 presents that when the laser pulse energy increases from 20 μ J to 73.6 μ J, the crater-like surface features become increasingly prominent, which is a common surface morphology in long-pulse or continuous laser processing. However, when the laser pulse energy is equal to or less than 10 μ J, the surface morphologies without any accumulation at the edge of the micropores are obtained (see Suppl. Mat. and Fig. 5(a)), which is the typical morphology of the short-pulse laser processing such

as femtosecond laser processing. This is because that on the one hand, for the low laser pulse energy ($\leq 10\mu\text{J}$), the deposition of laser energy on the surface of the materials is insufficient to trigger the thermal melting removal mechanism of the materials, and instead, the removal of the material is mainly achieved through non-thermal melting mechanisms. The energy absorption mechanism of the SiC is nonlinear absorption as a single photon cannot stimulate electrons from a valence band to a conduction band. The band gap of SiC is about 2.54eV [24,25], which is larger than that of a single photon. Multiphoton absorption mechanism, tunnel ionization absorption mechanism and avalanche ionization absorption mechanism are the three main mechanisms for the nonlinear absorption of SiC. Firstly, the electrons of the SiC absorb the energy of the multiple photons and transfer to a higher energy level as the combined energies of the multiple photons are sufficient to the energy level transition from the valence band to the conduction band. Secondly, the powerful electric field induced by the femtosecond laser can significantly change the coulomb potential energy of atoms. In the femtosecond laser processing, when the electric field intensity is high enough, the total potential barrier feel by the electrons will decrease or become thinner. The bound electrons can utilize the quantum tunneling effect to pass through the barrier. The electrons change from a bound state to a free state and become free electrons within a femtosecond. Thirdly, a small amount of initial free electrons are produced by multi-photon absorption and tunnel ionization mechanism for the avalanche ionization. With the action of the femtosecond laser field, these free electrons acquire kinetic energy. High-energy electrons collide with atoms, resulting in more free electrons. The newly produced electrons repeat this process, generating an exponential increase in the number of electrons. Finally, the electrons transfer the laser energy to phonons, and then the phonons deliver the laser energy to the lattice of SiC. However, the aluminum mainly undergoes a linear absorption process. Then, the laser energy is transferred sequentially through photons - electrons - phonons - lattice-continuous medium, and is transmitted to the nearby area directly irradiated by the femtosecond laser. When the laser pulse energy is low, the laser energy is largely dissipated at the phonon to lattice stage. The energy transfer process in the continuous medium has become extremely weak, and the thermal effect is not obvious. The material removal is basically achieved through non-thermal effects. The coulomb explosion, vaporization and plasma shock are the main mechanisms for the materials removal with low laser pulse energy.

On the other hand, for the large laser pulse energy ($>10\mu\text{J}$), the rim elevation observed around the crater is likely caused by recoil pressure from plasma expansion, which ejects partially molten nano-droplets outward, rather than by sustained thermal melting flow.

Fig. 6 displays the cross-section of the micropores with different laser pulse energy. It can be seen from the figure that when the laser pulse energy increases from $20\mu\text{J}$ to $73.6\mu\text{J}$, the height of the accumulated material at the crater-like edge gradually increase

from about $2\mu\text{m}$ to $8\mu\text{m}$. This is because as the laser pulse energy increases, the more energy is deposited into the materials, the more molten material there is. The height of the subsequent formed accumulation is increased due to the impact of the high-temperature and high-pressure plasma.

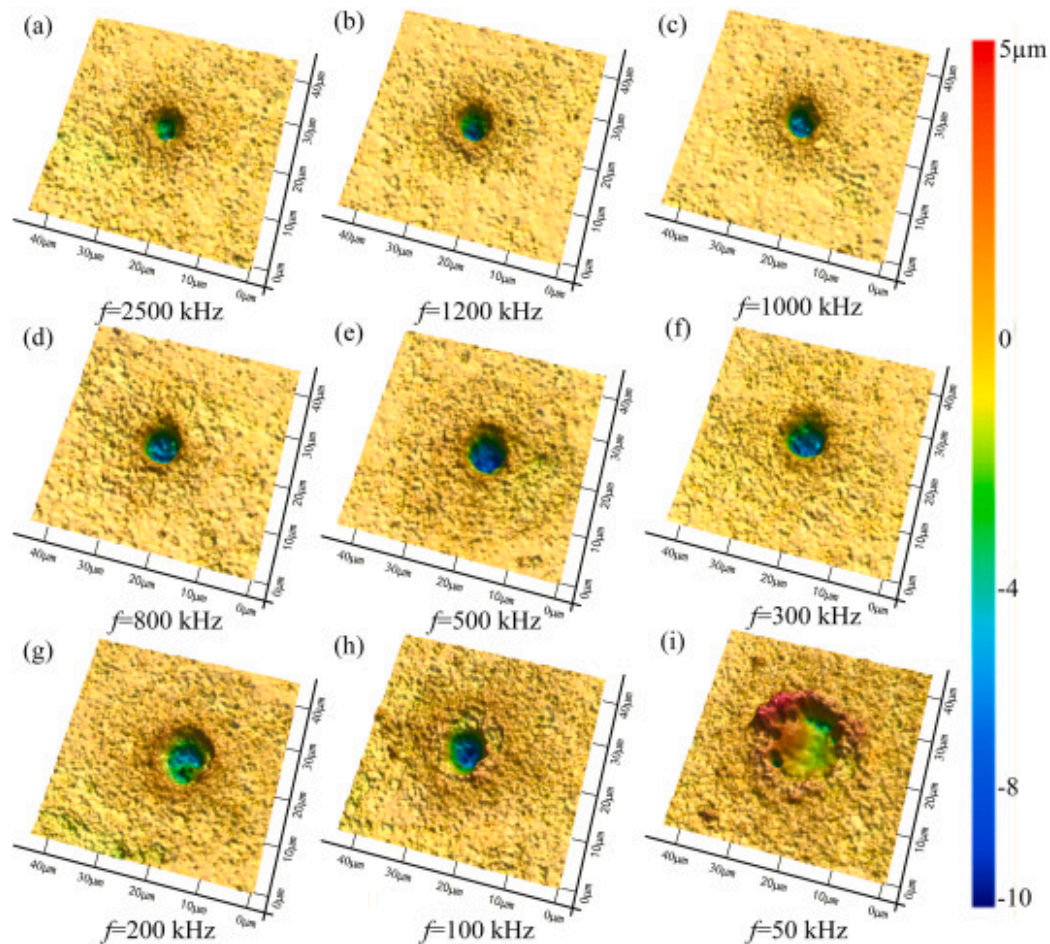


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Fig. 6. Cross-section of ablation micropores by femtosecond laser processing with different laser pulse energy (E): **(a)** $10\mu\text{J}$, **(b)** $20\mu\text{J}$, **(c)** $30\mu\text{J}$, **(d)** $35\mu\text{J}$, **(e)** $40\mu\text{J}$, **(f)** $45\mu\text{J}$, **(g)** $50\mu\text{J}$, **(h)** $60\mu\text{J}$ and **(i)** $73.6\mu\text{J}$.

The three-dimensional morphology of the ablation micropores by femtosecond laser processing with different repetition frequency (f) were exhibited in Fig. 7. On the right side of Fig. 7, the color bar is the dimension variation with value from $-10\mu\text{m}$ to $5\mu\text{m}$.



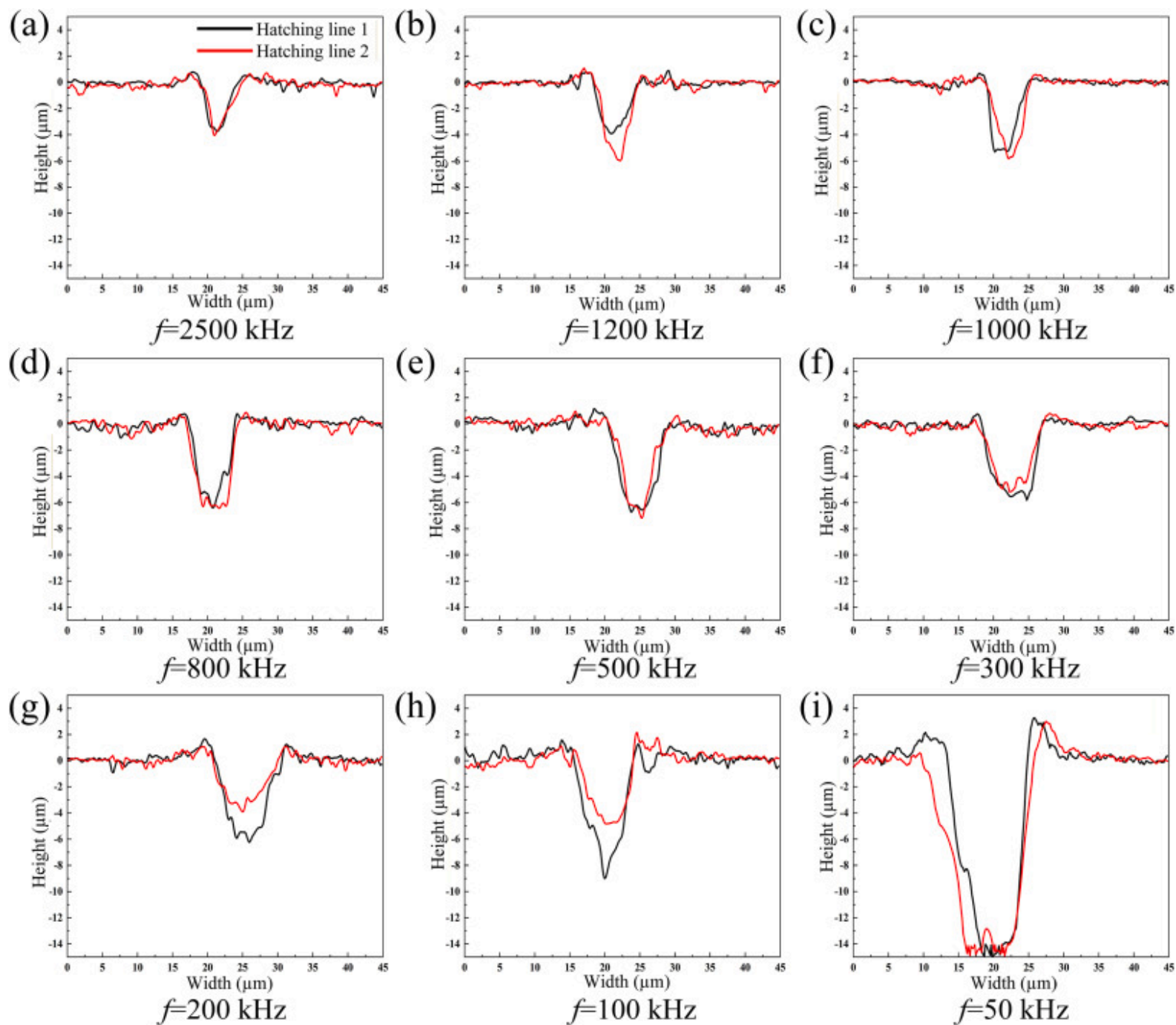
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Fig. 7. Three dimensions morphology of ablation micropores by femtosecond laser processing with different laser repetition frequency (f): **(a)** 2500kHz, **(b)** 1200kHz, **(c)** 1000kHz, **(d)** 800kHz, **(e)** 500kHz, **(f)** 300kHz, **(g)** 200kHz, **(h)** 100kHz and **(i)** 50kHz.

Fig. 7 presents that when the laser repetition frequency decreases from 2500kHz to 100kHz, the accumulation at the edge of the micropores are not obvious. However, when the laser repetition frequency decreases is 50kHz, the crater-like surface features is ablated. This is because that with low repetition frequency, high laser pulse energy is obtained. Thus, the deposition of laser energy on the materials is sufficient to induce the thermal melting. Then, the molten materials was impacted to the edge area of the micropore surface by the high-pressure laser-induced plasma, resulting in a volcanic crater-like surface morphology.

Fig. 8 exhibits the cross-section of the micropores with different laser repetition frequency. It can be seen from the figure that when the laser repetition frequency decreases from 2500kHz to 50kHz, the depth of micropore gradually increase from about 4 μ m to 14 μ m. This is because as the laser repetition frequency decreases, more energy is deposited into the materials. When the laser repetition frequency is 50kHz, the height of the subsequent formed accumulation is increased to about 2 μ m due to the impact of the laser-induced plasma.

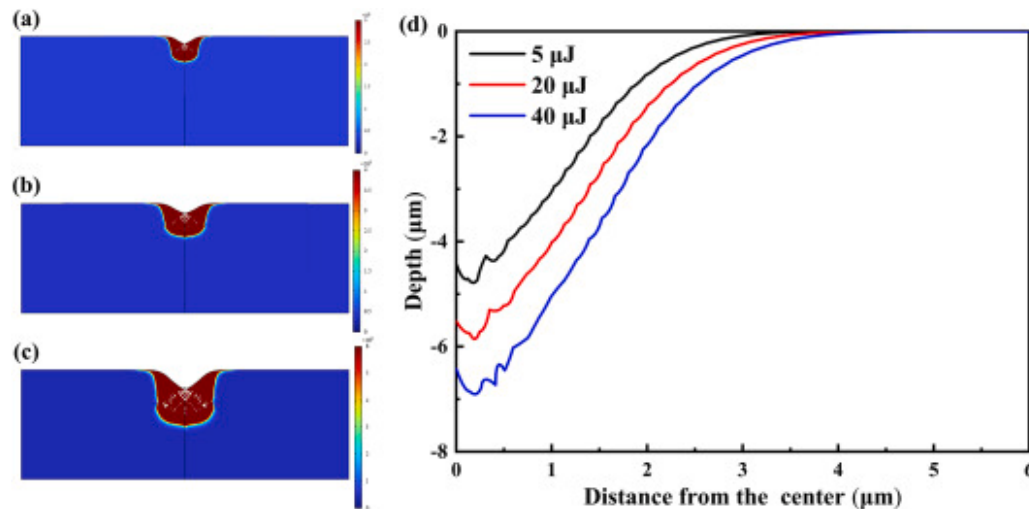


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Fig. 8. Cross-section of ablation micropores by femtosecond laser processing with different laser repetition frequency (f): **(a)** 2500kHz, **(b)** 1200kHz, **(c)** 1000kHz, **(d)** 800kHz, **(e)** 500kHz, **(f)** 300kHz, **(g)** 200kHz, **(h)** 100kHz and **(i)** 50kHz.

Fig. 9 presents the cross section and depth of the micropores by simulation. It can be seen from the figure that the depth of materials ablation continuously increases as the laser pulse energy rises (Fig. 9(a)–(b) and (c)). When the laser pulse energy increases from 5 μJ to 40 μJ , the ablation depth of the micropore increases from about 4.8 μm to 7 μm (Fig. 9(d)).

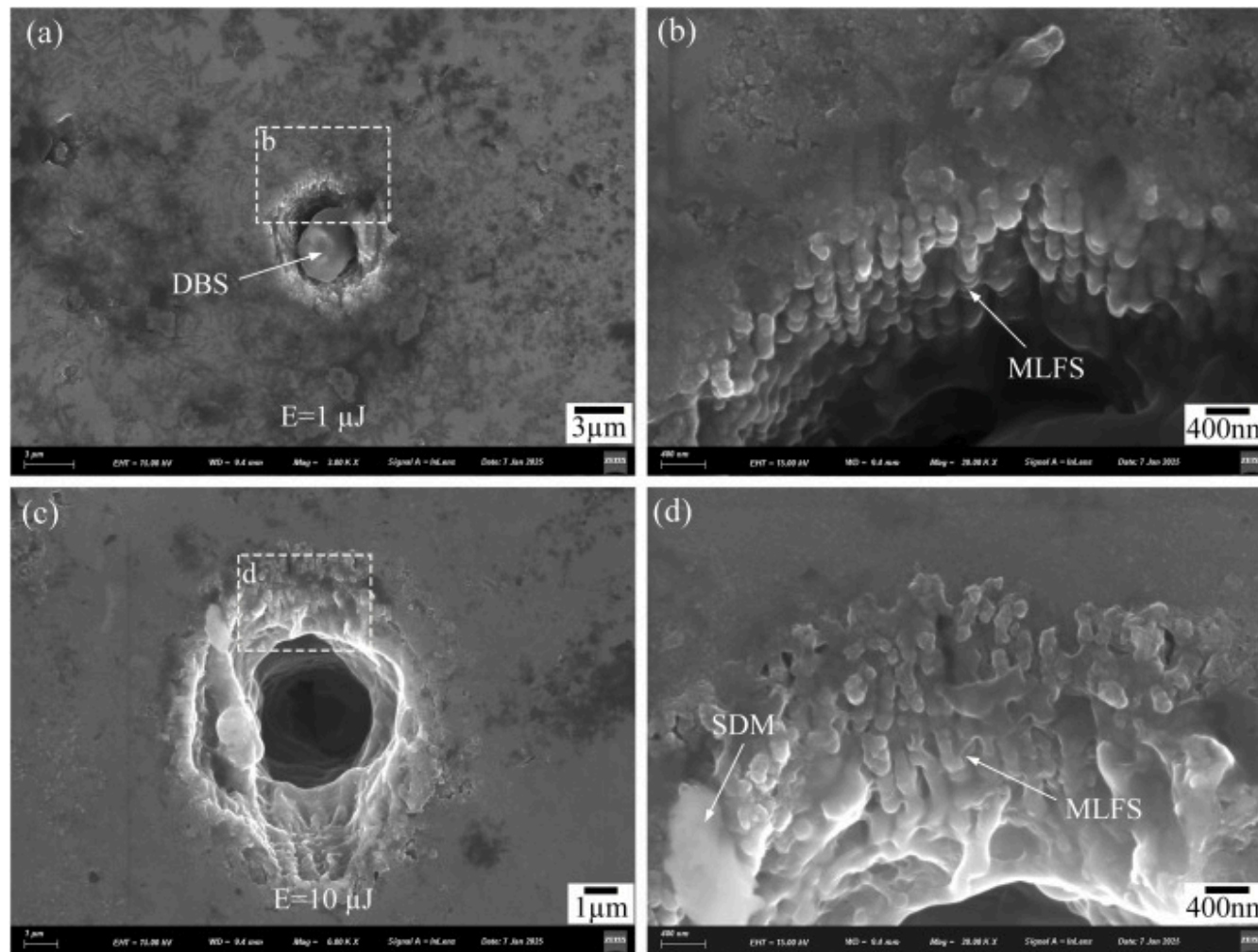


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Fig. 9. Cross section of the micropores by simulation: **(a)** 5 μJ , **(b)** 20 μJ , **(c)** 40 μJ and **(d)** depth of the micropores by simulation.

Fig. 10 exhibits the microstructure of the micropore on the surface of the SiCp/Al composites by femtosecond laser processing with laser pulse energy of 1 μJ and 10 μJ . No debris was found at the periphery of the micropore, but there were still detached block substances (DBS) at the center of the micropore (Fig. 10(a)). This is likely due to the insufficient laser energy (1 μJ), which was not strong enough to ablate the wide band gap and higher-melting-point SiC. The diameter of the micropore is about 4 μm , which is consistent with that value obtained by the laser scanning confocal microscope.



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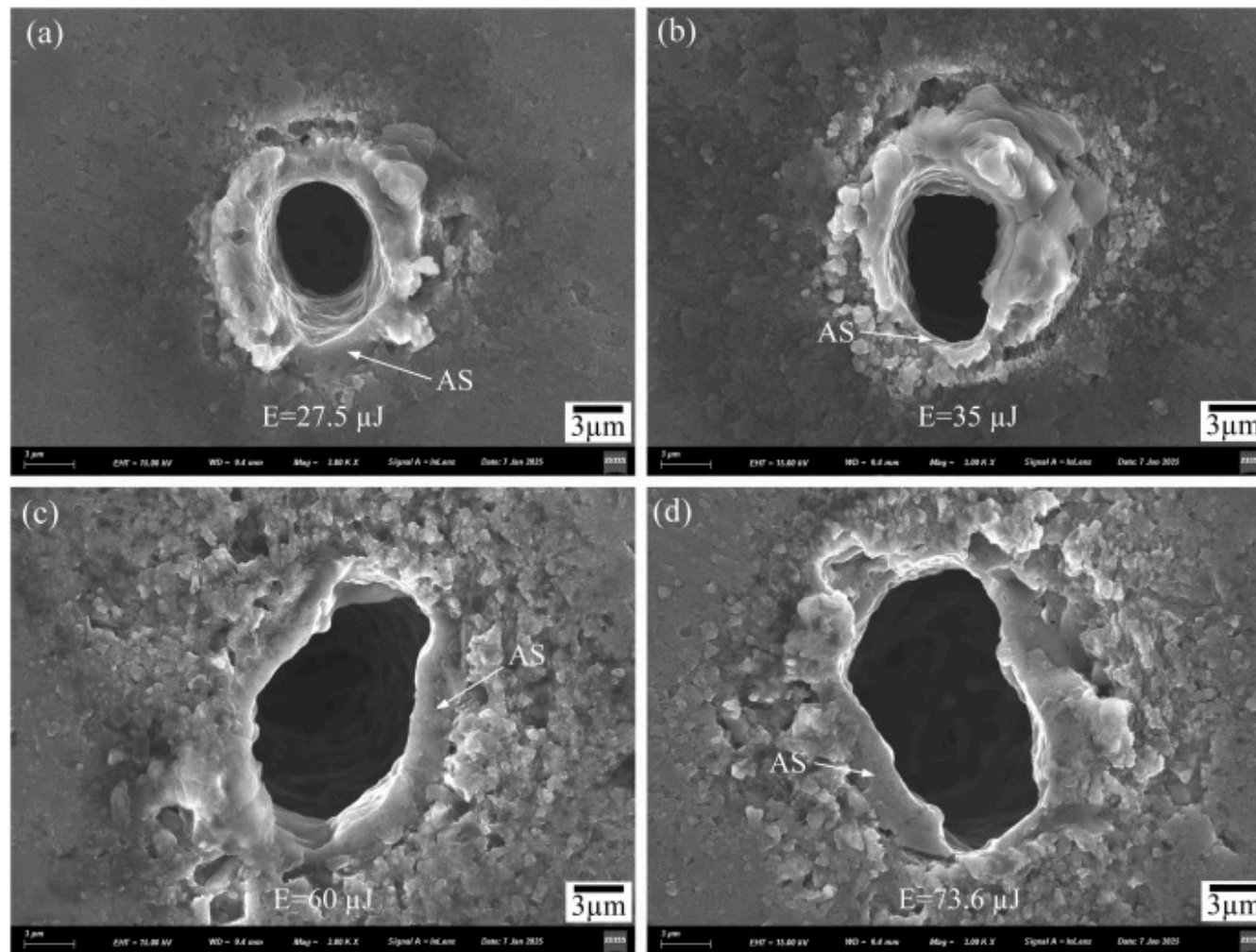
Fig. 10. Microstructure of the ablation micropores by femtosecond laser processing with: **(a)** and **(b)** 1 μJ , **(c)** and **(d)** 10 μJ .

This research result was consistent with that of Zhang et al. [19]. Their research reported that the Al matrix was removed before the SiC particles. Zhang et al. also indicated that the Al area and SiC area in the SiCp/Al composites were concave and convex, respectively, due to the removal mechanism of vaporization in the femtosecond laser processing of SiCp/Al composites. It can be

observed that there are a lot of multi-layer fish-scale periodic microstructures (MLFS) on the inner walls around the micropores (Fig. 10(b)). This is likely because that when the femtosecond laser acts on the SiCp/Al composites, the materials is layer by layer subjected to coulomb explosion, vaporization and plasma shock ablation, thus leaving a multi-layered periodic microstructure on the inner walls of the micropores.

Fig. 10(b) exhibits the microstructure of the micropore on the surface of the SiCp/Al composites by femtosecond laser processing with 10 μ J. Comparing with the micropores ablated by 1 μ J, the micropores formed by 10 μ J has poorer roundness and is elliptical in shape. This is because that during the layer-by-layer ablating process of the plasma shock, since silicon carbide and aluminum matrix are randomly fused together, at certain locations, there are block-like silicon carbide at the edge of the pores. Therefore, when the micropore is ablated off, a notch appears at that spot, resulting in the elliptical micropores. Furthermore, within the micropores, there are still layers of periodic microstructures resembling fish scales, accompanied by relatively obvious melting marks in some areas. This is because that with 10 μ J of laser pulse energy, more the laser energy is deposited and transferred to the crystal lattice and continuous medium of the materials, resulting in local melting. Shaped delaminated materials (SDM) is found at the edges of the micropore, which is caused by the plasma shock. This results are different from that of Pan et al. [26]. In their research, the LIPSS on the surface of SiC composite was obtained without micropores. This might be the low laser pulse energy in their research. Additionally, Gemini et al. [27] and Dong et al. [28] reported that LIPSS was ablated on the surfaces of the materials due to the periodic energy deposition in the femtosecond laser processing. The interference action between the incident femtosecond laser beam and the scattered electromagnetic waves was the main reason for this LIPSS mechanism.

Fig. 11 presents the microstructure of the micropore on the surface of the SiCp/Al composites by femtosecond laser processing with laser pulse energy from 27.5 μ J to 73.6 μ J. It can be seen from Fig. 11 that as the laser pulse energy increases from 27.5 μ J to 73.6 μ J, the diameter of the microholes gradually increases, while the roundness of the microholes gradually deteriorates. This is because that the greater the laser pulse energy, the greater the effects of coulomb explosion, vaporization and plasma shock, and more materials are removed away. Furthermore, when the laser energy was 60 μ J and 73.6 μ J, very distinct ablating steps (AS) appeared at the edges of the micropores, resembling the disintegration fracture surface morphology of the brittle fracture of the materials. This phenomenon indicates that the greater the laser energy, the more likely the deposited laser energy is to cause stronger recoil pressure from plasma expansion. The materials are stripped off in chunks, thus leaving the ablated steps.



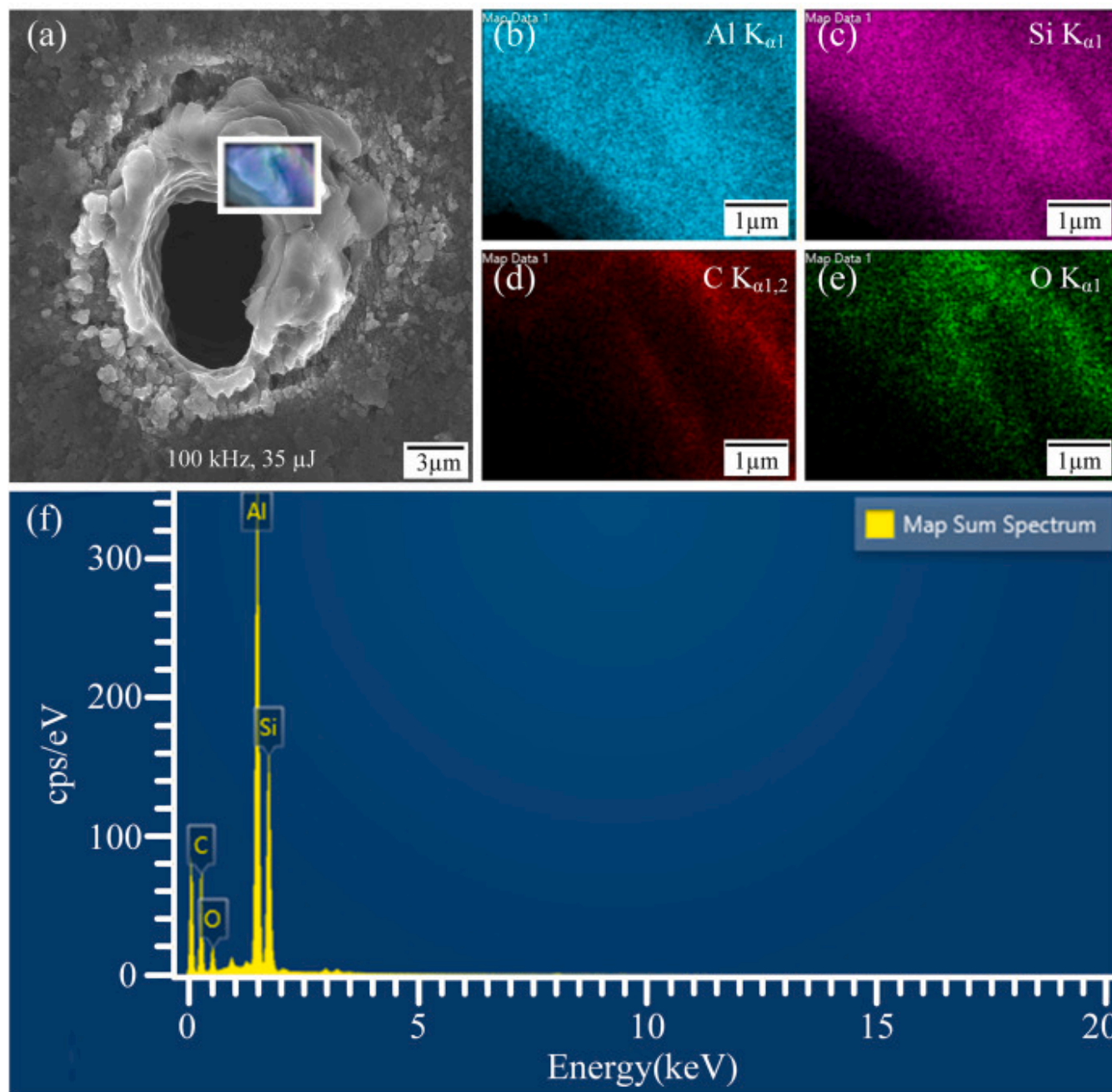
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Fig. 11. Microstructure of the ablation micropores by femtosecond laser processing with: (a) 27.5 μJ , (b) 35 μJ , (c) 60 μJ and (d) 73.6 μJ .

In order to analyze the composition of the debris at the edge of the micropores, an EDS energy spectrum surface scan was conducted on the debris. Fig. 12 shows the distribution of various elements in the energy spectrum scan of the edge debris of the

micropores by femtosecond laser processing with a laser pulse energy of $35\mu\text{J}$. It can be seen from the figure that the debris is mainly composed of Al, Si, C and O. This indicates that due to the large amount of laser energy deposition, the thermal effect induces a significant oxidation phenomenon at this location [29].



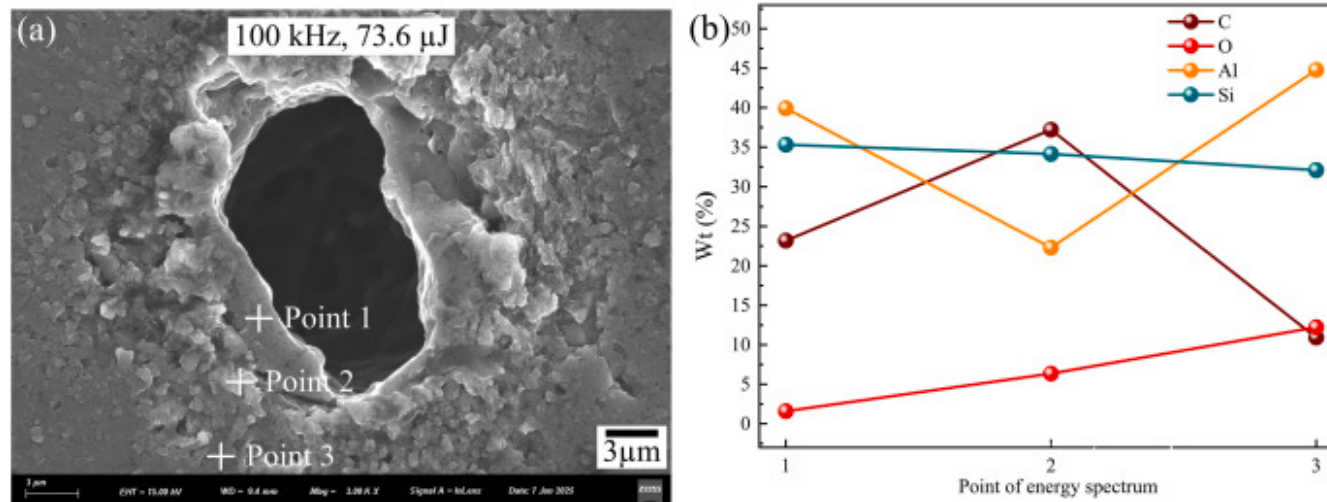
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Fig. 12. Energy dispersive spectrum (EDS) of the ablation micropores by femtosecond laser processing with laser pulse energy of $35\mu\text{J}$: **(a)** scanning area, **(b)** Al element, **(c)** Si element, **(d)** C element, **(e)** O element and **(f)** EDS.

The oxidation process in SiCp/Al composite ablation by femtosecond laser is rather complex, involving passive oxidation and active oxidation of silicon carbide, volatilization of silicon dioxide and reaction decomposition of silicon carbide and silicon dioxide, etc. [30]. The active oxidation of silicon carbide requires a reaction with oxygen in the atmosphere at high temperatures. When femtosecond laser is applied to the surface of SiCp/Al composite materials, the central temperature of the material gradually increases. Gao et al. [31] also indicated that for silicon carbide composite materials, the ablation area of femtosecond laser pulses in the center region of the material presents unoxidized areas through the passive oxidation mechanism, with a smooth surface. While around the center region, an oxidized area containing a large number of oxide particles were obtained through the active oxidation mechanism.

Fig. 13 displays the point analysis of EDS of the ablation micropores by femtosecond laser processing with laser pulse energy of $73.6\mu\text{J}$. It can be seen from the figure that at the step of the micropores (point 1), there is almost no O element, indicating that this area has not been oxidized significantly. However, at the position of point 2 and point 3, the content of the oxygen element gradually increases, suggesting that the more close to the outer edge of the micropores, the more severe the oxidation phenomenon becomes. Additionally, the Al content is relatively high at all three locations. This is because that due to the high energy of the laser, the thermal effect is significant. The Al accumulates and solidifies at the edge of the micropores due to the impact of the plasma shock, resulting in a higher content of Al element. This result is consistent with the previous ones obtained by the laser pulse energy of $35\mu\text{J}$.



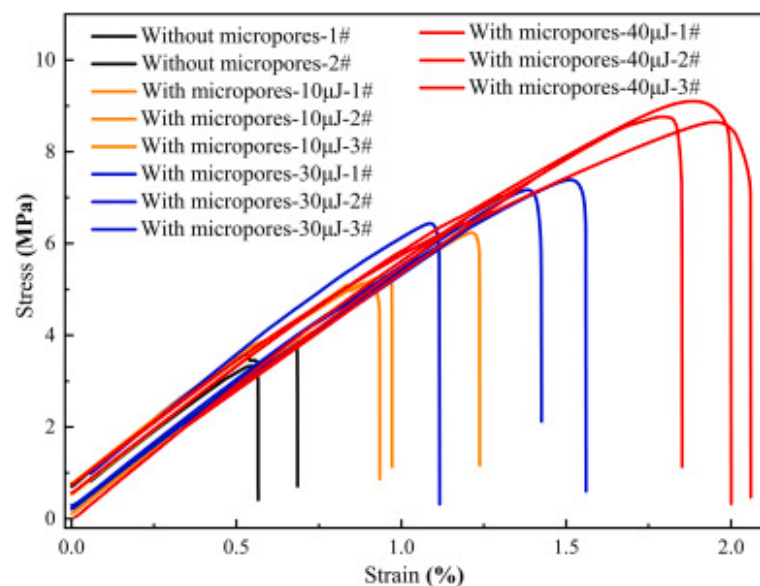
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Fig. 13. Point analysis of EDS of the ablation micropores by femtosecond laser processing with laser pulse energy of 73.6 μJ: **(a)** position of the points and **(b)** Wt (%) of the elements.

3.3. Adhesive bonding joint strengthening mechanism

Fig. 14 exhibits stress-strain curves of the bonded joints without and with micropores array by femtosecond laser processing with 10 μJ, 30 μJ and 40 μJ.



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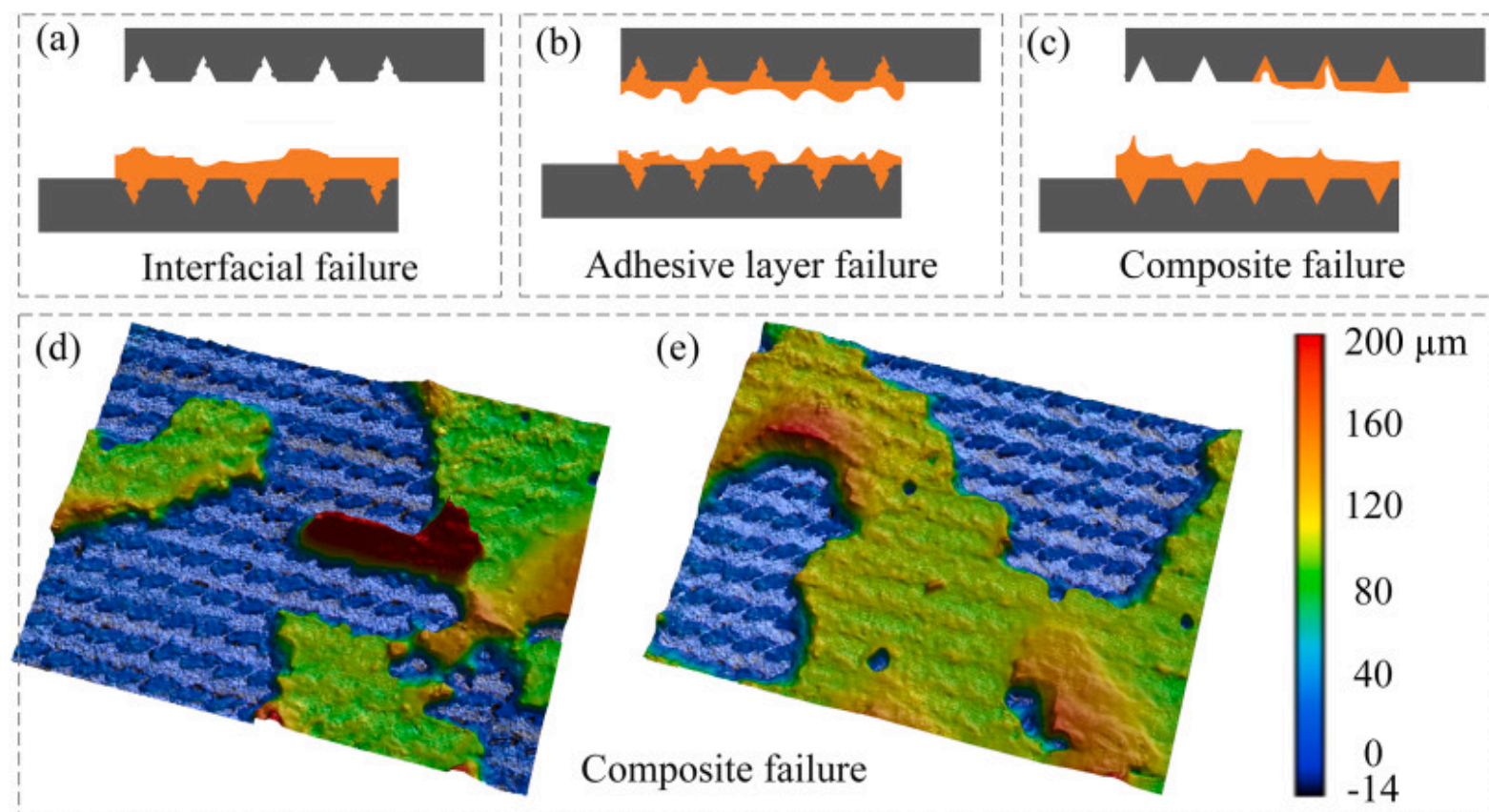
Fig. 14. Stress-strain curves of the bonded joints without and with micropores array by femtosecond laser processing with 10μJ, 30μJ and 40μJ.

It can be seen from the figure that the tensile strength of the adhesive joints without the micropores array by femtosecond laser processing is only 3.5MPa, while the values of the tensile strength of the adhesive joints with the micropores array are all higher than that of the specimens without micropores array. The tensile strength of the adhesive joints gradually increases as the laser pulse energy increases. When the laser energy increase from 10μJ to 40μJ, the tensile strength of the adhesive joints rose from approximately 5.2MPa–8.5MPa. The maximum tensile strength of the bonded joint with microstructure array is approximately 3 times that of the unprocessed specimens. Additionally, the average elongation of the adhesive joints with micropores array by femtosecond laser processing with laser pulse energy of 40μJ is approximately 1.8%, which is about 3 times that of the adhesive joints without micropores array.

The stress-strain results of the adhesive joints show that the strength and elongation of the adhesive joints with micropores have been significantly improved. This might be because that for the adhesive joints with micropores arrays, the adhesive can

penetrate into the micropores to form adhesive pins. The adhesive pins can generate a pinching effect and realize the mechanical interlocking mechanism [32]. When the adhesive joints are subjected to destructive force, which can enhance the tensile strength and elongation of the adhesive joint. The greater the laser pulse energy, the larger the diameter and depth of the processed micropores, and the more obvious the pinning effect. Thus, the tensile strength and elongation gradually increase with the increase of laser pulse energy.

Fig. 15 presents the schematic diagram of the three typical failure cross-section of adhesive joints such as interfacial failure, adhesive layer failure and composite failure, and the failure surface morphologies of the adhesive joint by femtosecond laser processing with laser pulse energy of $40\mu\text{J}$ (2#). It can be seen from the figure that due to the presence of a layered fish-scale periodic microstructure inside the micropores, the bonding between the adhesive pins and the micropores is relatively strong. Complete interface failure has not been occurred in the specimens with micropores array. Meanwhile, since the adhesive used is aviation glue, its bonding performance is also excellent. There was no failure of the adhesive layer. The adhesive joint fails in the form of a composite failure.



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Fig. 15. Schematic diagram of the three typical failure cross-section of adhesive bonding joints: **(a)** interfacial failure, **(b)** adhesive layer failure and **(c)** composite failure, **(d)** and **(e)** the failure surface morphology of the adhesive bonding joint with laser pulse energy of $40\mu\text{J}$ (2#).

4. Conclusion

The primary motivation for this study was to reveal the materials removal and adhesive bonding joint strengthening mechanisms of aluminum-based silicon carbide metal matrix composites by femtosecond laser processing. Based on the results and discussion, the following conclusions were obtained.

- (1) The materials removal mechanism of SiC_p/Al composites by femtosecond laser processing was layer-by-layer materials ablation. When the laser pulse energy was equal or less than 10μJ, periodic microstructure of multi-layer fish scales with length of about 500nm were obtained in the inner wall of the micropore due to the non-thermal coupling effect of coulomb explosion, vaporization and plasma shock. When the laser pulse energy exceeded 10μJ, brittle fracture steps with width of about 2.5μm were obtained at the edge of the volcanic crater-shaped micropore mainly due to the recoil pressure from plasma expansion. The laser ablation threshold of the SiC_p/Al composites by the femtosecond laser processing with repetition frequency of 100kHz was 2.91J/cm².
- (2) The tensile strength and elongation of the adhesive bonding joints with micropores array have been significantly improved due to the pinning effect with mechanical interlocking mechanism by the micropores. The tensile strength and elongation of the adhesive bonding joints gradually increased when the laser pulse energy increased because of that the pinning effect becomes more pronounced as the diameter and depth of the micropores increased with the increasing of laser pulse energy. The average tensile strength and elongation of the adhesive bonding joints with micropores array by laser pulse energy of 40μJ were about 8.5MPa and 1.8%, respectively, which are about 3 times those of the joints without micropores array with value of 2.5MPa and 0.6%, respectively.
- (3) The adhesive bonding joints with micropores array failed in the form of a composite failure including of interfacial failure and adhesive layer failure as the multi-layer fish-scale periodic microstructure on the inner wall of the micropores and the strengthening effect of the volcanic crater-like deposits at the micropore edges increased the adhesive bonding area, enhancing the pinning effect with mechanical interlocking mechanism and significantly preventing a pure interface failure.

CRedit authorship contribution statement

Jiecai Feng: Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Hengyi Song:** Investigation. **Xing Wang:** Writing – original draft, Investigation, Data curation. **Hongfei Liu:**

Formal analysis. **Yanning Sun:** Formal analysis. **Xiaohai Peng:** Data curation. **Yilian Zhang:** Investigation. **Yingzhong Tian:** Methodology.

Declaration of generative AI

No artificial intelligence was used in the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Key R&D Program of China [[2023YFB3307700](#)] and the Aeronautical Science Foundation of China [[202400540S6002](#)].

Appendix A. Supplementary data

The following is the supplementary data to this article:

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Multimedia component 1.

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